

COLLABORATOR SIDE



Collaborator X_j + LiDAR

PointPillars / BEV backbone

shared weights

Object-level message L

Feature codec

Feature-level message F (16-QAM)

≈ 0.024 Msym/frame

≈ 0.99 Msym/frame

C256: 256-QAM comparator (not deployed), ≈ 0.495 Msym/frame

dominated; excluded from deployment
(§III-B)

select one, controlled by s

(one message)

2-bit request (via CAM)

EGO SIDE — ON-VEHICLE DECISION



Ego vehicle X_e

“Selector runs on the ego vehicle”

LiDAR

PointPillars / BEV backbone

shared weights

Cue Extractor

21-d

Channel estimator

(SNR, dB)
(AWGN/ Rayleigh)

RF Selector

$g(z_t, \hat{y}_t, \hat{\sigma}_t) \mapsto s_t$

Fusion Ψ

$F_{ego,t} + m_{s_t}$

Detection head D

(3D boxes)

V2V wireless channel (SNR, BLER)

local detection output

$F_{ego,t}$

request: ego $\rightarrow$ collaborator message: collaborator $\rightarrow$ ego